



AI Accountability Policy Request for Comment: Ada Lovelace Institute

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AI Accountability Objectives

- Definitions of accountability practices and assessments
- Definitions of normative concepts
- What is the purpose of these assessments?
 - What are they set up to achieve?
 - What are their goals?
- What is their potential scope?
 - What should they contain?
 - How widely or narrowly should they understand risk, bias, social justice etc?

Definitions for accountability practices:

Meaningful AI accountability practices understand accountability as a relational concept between actors and a forum: the actor has an obligation to explain/ justify conduct, and the forum can pose questions, pass judgment, and enforce consequences against the actor.¹

There are a range of different accountability practices in the AI sector. The Ada Lovelace Institute has led work around defining key terms in the algorithm accountability space: for example, *Examining the Black Box*, outlines different methods

¹ Mark Bovens (2007) Analysing and assessing accountability: a conceptual framework. Available at: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0386.2007.00378.x>



for assessing algorithmic systems and their definitions² as well as synthesising emerging accountability policy mechanisms, including assessments and audits, in a partnership with the AI Now Institute and the Open Government Partnership³. We also conducted a survey of six technical methods for auditing algorithmic systems, defining key methodologies⁴.

By assessment, we consider these to be evaluations of the potential and actual impacts that a technology has on people and society. These include practices like Algorithmic Impact Assessments (AIAs) or risk assessments, which are usually conducted in early stages of the design process of a technology to identify and mitigate the potential harmful impacts of a system (and crucially, to document those impacts), and can also take place after a system is in place to understand the actual impacts of a system. It also includes algorithmic auditing practices, with seek to provide a technical evaluation of an AI system's performance and biases before and after it has been deployed.

. Accountability assessments of AI systems must take a broad and sociotechnical lens that evaluates not just the technology, but the people, institutions, environments and relationships that shape how that technology is used and understood. Accountability assessments must therefore home in on the institutions designing, developing, procuring and deploying the technology, not just the technology itself. It is important to emphasise, however, that the presence of a particular assessment alone does not constitute a forum to which a developer or user of an AI system can be held accountable⁵.

Under this concept of accountability, assessments are themselves not a form of accountability – to enable questions, judgment, and enforcement of sanctions, assessments must be documented (such as in a published AIA or audit), made available to the forum (be that the public or a regulator), and action taken on them (for example, the developer revoking the use of a tool that is likely to cause harm). It is also important that people affected by development and use of AI systems should be given a stake in

² Ada Lovelace Institute and DataKind UK (2020) Examining the black box. Available at: <https://www.adalovelaceinstitute.org/wp-content/uploads/2020/04/Ada-Lovelace-Institute-DataKind-UK-Examining-the-Black-Box-Report-2020.pdf>

³ Ada Lovelace Institute, AI Now Institute and Open Government Partnership (2021) Algorithmic accountability for the public sector. Available at: <https://www.adalovelaceinstitute.org/report/algorithmic-accountability-public-sector/>

⁴ Ada Lovelace Institute (2022) Technical methods for regulatory inspection of algorithmic systems. Available at: <https://www.adalovelaceinstitute.org/report/technical-methods-regulatory-inspection/>

⁵ Ada Lovelace Institute (2022) Technical methods for regulatory inspection of algorithmic systems. Available at: <https://www.adalovelaceinstitute.org/report/technical-methods-regulatory-inspection/>



defining values for their development, use and governance across different applications and sectors – see for example the Ada Lovelace Institute’s Citizens Biometrics Council⁶.

Motivations:

There may be multiple underlying motivations or explicitly defined goals for an assessment, and each approach may bring both opportunities and limitations.

The Ada Lovelace Institute understands the primary function of audits and assessments is to make developers and deployers of AI systems accountable to those affected by them⁷, but others may use assessments to clarify and standardise internal processes by providing a clear and coherent methodological framework to follow: Microsoft’s Responsible AI impact assessment follows this approach⁸. Our research project designing an algorithmic impact assessments in healthcare with the UK National Health Service (NHS)’s AI Lab designed a AIA process for a data access context, where a completed AIA artefact helps create an accountability relationship between people affected by AI in healthcare and the developers of tools using their data, and supports the objectives of the NHS of ‘maximising the benefits and minimising the risks of healthcare’⁹. A publicly available artefact also enables greater levels of transparency for people who are impacted by this tool to understand how it will affect them.

No single accountability mechanism offers a complete solution to assess the risks of AI systems: individual assessments should be considered as part of a broader toolkit of practices. The UK’s Centre for Data Ethics and Innovation (CDEI)’s AI assurance toolkit offers an example of a set of AI assessment practices throughout the lifecycle of an AI system’s development and deployment¹⁰. However, adopting a broader, systemic lens to assessment enables a more holistic understanding of potential impacts. Our AIA process (see [AIA template](#)) offers an example of an accountability mechanism that takes a systemic lens to risk and harm both within and between communities. Prompts ask teams completing the AIA to consider, for example, consent and autonomy across certain groups, and potential environmental impacts of the proposed AI system. Other impact assessments, such as the Institute for the Future of Work (IFOW) AIA specifically

⁶ Ada Lovelace Institute (2021) The Citizen’s Biometrics Council. Available at:

<https://www.adalovelaceinstitute.org/project/citizens-biometrics-council/>

⁷ Ada Lovelace Institute (n.d) Programme: Ethics & Accountability in Practice. Available at:

<https://www.adalovelaceinstitute.org/our-work/programmes/ethics-accountability-practice/>

⁸ Microsoft (2022) Responsible AI: Impact Assessment Guide. Available at: <https://blogs.microsoft.com/wp-content/uploads/prod/sites/5/2022/06/Microsoft-RAI-Impact-Assessment-Guide.pdf>

⁹ Ada Lovelace Institute (2022) Algorithmic impact assessment: a case study in healthcare. Available at:

<https://www.adalovelaceinstitute.org/project/algorithmic-impact-assessment-healthcare/>

¹⁰ Centre for Data Ethics & Innovation (CDEI) (n.d) AI assurance guide (BETA) Available at:

<https://cdeiu.github.io/ai-assurance-guide/>



focus on workers and AI impacts in the workplace as target populations/impacts¹¹. Emerging legislation, such as the New York City Local Law 144 (known as the Bias Audit Law)¹² states that bias audits must require examination of outputs of an automated employment decision tool (AEDT) to explore bias along single categories of identity, e.g race, gender, and at the intersection of categories, e.g race and gender together.

Existing Resources and Models

- What AI accountability mechanisms are currently being used?
- Are the accountability frameworks of certain sectors, industries, or market participants especially mature as compared to others?

Existing algorithmic accountability mechanisms:

In addition to prominent accountability mechanisms currently mandated in regulation/legislation, such as the Canadian government's Directive on Automated Decision Making's AIA, and the independent audit requirements under the Digital Services Act, we draw attention to current UK initiatives:

UK initiatives for consideration

- UK Algorithmic Transparency Recording Standards: developed by the Central Digital and Data Office and Centre for Data Ethics and Innovation, the framework provides an algorithmic transparency standard for use by public sector organisations across the UK (sector agnostic).¹³ This standard provides a range of information about AI systems, including the data used to train them.
- AI Standards Hub: led by the Alan Turing Institute in partnership with the British Standards Institute (BSI) and the National Physical Laboratory (NPL), the Hub provides an AI-related standards and a policy database, collating standards for different AI applications.¹⁴ Several of these standards may touch on the development of accountability assessment practices like auditing and evaluations.
- UK ICO AI and Data Protection Risk Toolkit: aims to combine data protection regimes with particular considerations for AI, and takes the form of a

¹¹ IFOW (2021) Algorithmic impact assessments. Available at: <https://www.ifow.org/publications/policy-briefing-building-a-systematic-framework-of-accountability-for-algorithmic-decision-making>

¹² New York City Council (2021) Local Law 144 Legislation Text. Available at: <https://aboutblaw.com/0vz>

¹³ Central Digital and Data Office (CDDO) and Centre for Data Ethics and Innovation (CDEI) Algorithmic Transparency Reporting Standard. Available at: <https://www.gov.uk/government/publications/algorithmic-transparency-template>

¹⁴ The Alan Turing Institute, BSI and NPL (2022) AI Standards Hub. Available at: <https://aistandardshub.org/>



spreadsheet of prompts related to UK GDPR and data protection risks at each stage of the AI lifecycle, with accompanying guidance, action recommendations and space for documenting the process¹⁵

- UK AI White Paper, 'A pro-innovation approach to AI regulation', which adopts an 'Accountability and governance' principle¹⁶, recommending use of algorithmic accountability measures including impact assessments and audits (though not statutorily required).
- *[Forthcoming]* UK NHS algorithmic impact assessment in healthcare: our AIA has been planned for pilot in a data access context (and has also been explored for use in other contexts¹⁷).

Accountability mechanisms from other sectors

The long history of environmental impact assessments (emerging under the US NEPA) in policy offers learnings for the potential for impact assessments for AI: frameworks for EIAs mandate public participation 'by design' to improve the legitimacy and quality of the EIA and to contribute to normative goals like democratic decision-making¹⁸. By contrast, data protection impact assessments generally do not make recommendations about incorporating a participatory component, nor is publishing the DPIA deemed a requirement. Accountability practices should look to EIA components for best practice in mandating participatory components.

As put forward by Raji et al. (2022)¹⁹, third-party reporting systems in certain sectors, for example, the FDA's medical device incident database or the Federal Aviation Administration's Aviation Safety Reporting System, function as effective post-market audit mechanisms and may be instructive for the AI context.

Accountability Subjects

- Where in value chain accountability mechanisms should be used?
- How should efforts be coordinated/communicated?
- How should we consider the commercial vs public procurement processes?

¹⁵ ICO (n.d) AI and data protection risk toolkit. Available at: <https://ico.org.uk/for-organisations/guide-to-data-protection/key-dp-themes/guidance-on-ai-and-data-protection/ai-and-data-protection-risk-toolkit/>

¹⁶ Department for Science, Innovation and Technology (DSIT) and Office for Artificial Intelligence (OAI) A pro-innovation approach to AI regulation. Available at: <https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach/white-paper>

¹⁷ Wright et al. (2023) Implementing algorithmic governance: clarifying impact assessments through mock exercises. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4349890

¹⁸ Glucker et al. (2013) Public participation in environmental impact assessment: why, who and how? <https://www.sciencedirect.com/science/article/abs/pii/S0195925513000711>

¹⁹ Raji et al. (2022) Outsider oversight: designing a third-party audit ecosystem for AI Governance. Available at: <https://arxiv.org/abs/2206.04737>



- Accommodating downstream unknowns?
- Scoping assessments according to the risk of the technology and or the deployment context?

Accommodating the diffuse and iterative development cycles for AI systems requires processes and mechanisms that can enable 'end to end' accountability. Accountability practices must occur throughout the lifecycle of an AI system, from early ideation and problem formulation to post-deployment. For example, you might layer a DPIA or a datasheet²⁰ at the design phase, an internal audit at testing, and an audit by a third-party at (re)deployment.

Relevant stakeholders might wish to communicate about the different stages of accountability efforts via a 'decision provenance framework'²¹, which helps to map chains of inputs and flow-on effects throughout systems. The artefact or 'result' of the assessment should also clearly articulate the intended goals of the assessment, as well as the limitations – what the process is set up to achieve, as well as what it is *not* set up to achieve, for the benefit of all stakeholders.

When considering how to present technical information in a manner that is legible to a more general audience, an approach similar to the UK Algorithmic Transparency Reporting Standard may be appropriate, in which 'two tiers' of transparency are adopted²¹: Tier 1 provides basic description of the algorithmic system and a contact email and is aimed at members of the public, while Tier 2 offers more detailed information regarding procurement, scope, decision-making processes intended for researchers and journalists.

The incentive structures for embedding or procuring audits/assessments for public vs private actors are different. As in the Canadian AIA, responsible procurement management of AI systems into public sector decision-making is incentivised both through legal mandates and soft commitments to principles of good governance. Our NHS AIA is not designed for procurement contexts, but offers evidence in how to embed an assessment across a public/private sector intersection, where assessments produced

²⁰ Gebru et al. (2018) Datasheets for datasets. Available at: <https://arxiv.org/abs/1803.09010>

²¹ Central Digital & Data Office (CDDO) and Centre for Data Ethics & Innovation (CDEI)(2023) Algorithmic transparency recording standard - guidance for public sector bodies. Available at: <https://www.gov.uk/government/publications/guidance-for-organisations-using-the-algorithmic-transparency-recording-standard/algorithmic-transparency-recording-standard-guidance-for-public-sector-bodies#step-2-complete-the-algorithmic-transparency-report-tier-1>



by private research labs need to be scrutinised and assessed by an independent body: an NHS data access committee, which acts as a source of institutional legitimacy²².

It is important that assessments are iterative: they should be undertaken at a regular cadence to capture new learnings and evidence. For example, we recommend that our NHS AIA is recompleted every two years by the organisation developing an AI system. A similar process would be more complicated to replicate across a vendor-customer pipeline, as well as for systems which are likely to have diverse downstream applications, such as large language models (LLMs). Mökander et al. (2023)²³ offer a three-layered framework for auditing, focused on a governance audit of the tech provider, a model audit of the model and a downstream application audit. Communication between the three 'layers' is integral in this framework, for example system outputs should inform ongoing model iteration.

Finally, building or adopting accountability mechanisms requires groundwork: Ada's AIA for healthcare required an understanding of the types of application of AI in healthcare and the existing regulatory context. The resulting process provides sector-specific prompts for organisations developing AI systems, and is not duplicative of existing regulatory frameworks or steps. For example, in the UK, the Medicines and Healthcare products Regulatory Agency (MHRA) requires that AI products for the healthcare market will need to be regulated as a medical device, whereby they receive a risk classification: medium, high etc. This may be true of other sectors e.g manufacturing that are already well-regulated. It is also important to note that assigning a risk score to a system is a value-laden exercise: a potential failure mode is a myopic understanding of risk based on limited participation in the risk-scoring process²⁴.

Accountability Inputs and Transparency

- What are the obstacles to the flow of information necessary for AI accountability either within an organization or to outside examiners?
- Should there be standardised reporting?

²² Ada Lovelace Institute (2022) Algorithmic impact assessment: a case study in healthcare. Available at: <https://www.adalovelaceinstitute.org/wp-content/uploads/2022/02/Algorithmic-impact-assessment-a-case-study-in-healthcare.pdf>

²³ Mökander et al. (2023) Auditing large language models: a three-layered approach. Available at: <https://arxiv.org/pdf/2302.08500.pdf>

²⁴ Ada Lovelace Institute (2022) Algorithmic impact assessment: a case study in healthcare. Available at: <https://www.adalovelaceinstitute.org/wp-content/uploads/2022/02/Algorithmic-impact-assessment-a-case-study-in-healthcare.pdf>



Robust accountability assessment would require regulators or external evaluators to have access to three kinds of evidence: policies (e.g. product description documents or content moderation policies that determine a system’s intended behaviour), processes (such as information on the governance practices for the developer of that tool, which may include providing access to the people responsible for that system) and (algorithmic) outcomes (such as the final output or impact of that system on people, whether they use the system or not). Crucially, this does not just mean access to the source code of that system. Barriers for acquiring this evidence could arise at the vendor/company level, for example, in highly decentralised, large corporations where information on certain systems does not exist on a central repository. Regulators will need carved out powers to enable independent auditors to have access to this information, as well as capacity and resources to conduct audits.

Research indicates the best practice for both enabling accountability and transparency involves the results and artefacts of an assessment being published on a central repository, accessible to the public²⁵.

In our NHS AIA process we provide a methodology for ‘participatory algorithmic impact assessment’, which involves affected communities and patients being involved in the AIA process to deliberate on potential impacts of AI systems and to facilitate normative goals for assessments such as inclusion. In this process, these communities are already important stakeholders for the project and so should be kept in the loop with ongoing decision-making around reporting.

Barriers to Effective Accountability

- What are the significant barriers to effective AI accountability in the private sector?
- What are the potential strategies?
- Does a dearth of standards/benchmarks impact the uptake of audits and assessments?
- What is the role of intellectual property rights, terms of service, contractual obligations, or other legal entitlements in fostering or impeding a robust AI accountability ecosystem?

Barriers

²⁵ Ada Lovelace Institute (2022) Algorithmic impact assessment: a case study in healthcare. Available at: <https://www.adalovelaceinstitute.org/wp-content/uploads/2022/02/Algorithmic-impact-assessment-a-case-study-in-healthcare.pdf>



Many accountability practices face considerable challenges for adoption in the private sector without regulatory incentives or legal mandates. In some cases, these accountability mechanisms are actively undermined by industry practices.²⁶ Below, we set out some barriers to accountability practices that we have uncovered in our research.

Lack of external researcher/auditor access

As we identify in our *Technical methods for regulatory inspection* report, frequently researchers are not granted API access by private companies to conduct an audit. In some cases, regulators may lack the necessary powers to compel companies of interest to provide API access, or to continuously monitor a system's behaviour.

Risk of capture

Some auditing standardisation efforts are at risk of industry capture. Our research on inclusive standards-setting in AI governance in the EU highlights the issue of standards development bodies relying on technology companies for input and expertise, at the risk of excluding civil society organisations and other stakeholders.²⁷

No incentive to experiment with emerging AI accountability practices in lieu of external pressures

In our forthcoming report, *Exploring the role of public participation approaches in commercial AI*, we explore some of the barriers for private sector firms in embedding public participation approaches, as an emerging AI accountability mechanism. We find that, at present, there is limited internal incentives to conduct participation (no executive leadership, no ringfenced funding) as well as limited external pressure (from corporate shareholders, reputational pressure, regulation), which is often true of ethics/accountability measures more broadly where processes are unstandardised or poorly defined²⁸

Potential strategies

Enabling a marketplace of independent auditors

Legislation/regulation could help foster a new marketplace of independent auditors by giving regulator powers to mandate private sector organisations undertake

²⁶ Algorithm Watch (2021) AlgorithmWatch forced to shut down Instagram monitoring project after threats from Facebook'. Available at: <https://algorithmwatch.org/en/instagram-research-shut-down-by-facebook/>

²⁷ Ada Lovelace Institute (2023) Inclusive AI governance: civil society participation in standards development. Available at: <https://www.adalovelaceinstitute.org/wp-content/uploads/2023/03/Ada-Lovelace-Institute-Inclusive-AI-governance-Discussion-paper-March-2023.pdf>

²⁸ Costanza-Chock, Raji and Buolamwini (2022) Who audits the auditors? Recommendations from a field scan of the algorithmic auditing ecosystem. Available at: <https://dl.acm.org/doi/abs/10.1145/3531146.3533213>



independent audits from a pre-approved and vetted pool of agencies/auditors²⁹, as present in financial services for example.

As one recent paper that compares auditing practices in other fields to AI auditing shows, enabling a marketplace of independent auditors would require:³⁰

- The establishment of a national incident reporting system to enable third parties to file complaints about algorithmic systems (this would help identify what auditors should be auditing for, and who to audit).
- In the public sector context, the establishment of a central repository that displays what kinds of algorithmic systems are in use in a given sector or by a particular government agency, and the results of any audits
- The creation of a national AI auditing body that sets standards of practice, reporting requirements, and certification for independent algorithmic auditors, and oversees their behaviour.
- The establishment of training, standardization, and accreditation for third-party AI auditors, along with a body that establishes professional norms of practice
- The establishment of a board that determines access to data and other information required for auditors to perform their evaluations

Powers to compel platforms to provide data/access

Independent audits conducted by academic labs or civil society organisations often fail because of a lack of access – regulators will need powers to enforce this³¹. As our *Rethinking data and rebalancing power* report suggests, one way forward may be for regulators to have the power to issue ‘access to platform data’ mandates for independent researchers (similar to the EU DSA provisions), who can collect and analyse data about potential harms or societal trends under strict data protection and security conditions (for example minimising the type of data collected and with a clear data retention policy). This would lead to essential understanding and increased

²⁹ Ada Lovelace Institute (2021) Technical methods for regulatory inspection of algorithmic systems. Available at: https://www.adalovelaceinstitute.org/wp-content/uploads/2021/12/ADA_Technical-methods-regulatory-inspection_report.pdf

³⁰ Raji et al. (2022) Outsider oversight: designing a third party audit ecosystem for AI governance’ Available at: <http://arxiv.org/abs/2206.04737>

³¹ Ada Lovelace Institute (2021) Technical methods for regulatory inspection of algorithmic systems. https://www.adalovelaceinstitute.org/wp-content/uploads/2021/12/ADA_Technical-methods-regulatory-inspection_report.pdf



scrutiny, accountability and oversight over how proprietary algorithms function and to understand their impact at the individual, collective and societal level.

Role of legal entitlements

Intellectual property, trade secrets, terms of service, contractual obligations can limit access and analysis of AI systems by third parties such as researchers and impede their ability to independently review and evaluate the technology for safety, fairness, transparency. Intellectual property can create legal barriers to sharing data and performance parameters of AI systems leading to opaque results and the inability to contest and challenge the way they operate. The Uber/Ola case where a Dutch court rejected the companies' arguments that transparency around automated decision making would expose their trade secrets is instructive³² - when there is such strong information and power asymmetry intellectual property and trade secret arguments do not stand.

AI Accountability Policies

- Should AI accountability policies and/or regulation be sectoral or horizontal, or some combination of the two?
- Should AI accountability regulation, if any, focus on inputs to audits or assessments (e.g., documentation, data management, testing and validation), on increasing access to AI systems for auditors and researchers, on mandating accountability measures, and/or on some other aspect of the accountability ecosystem?

In our report, *Regulate to innovate*, we set out recommendations for AI regulation for the UK Government in line with its ambition for innovation: the development of new, clear regulations for AI; improved regulatory capacity and coordination; and improved transparency standards and accountability mechanisms³³. We make the case for establishing a central function to oversee development and implementation of AI-specific, domain-neutral statutory rules for AI systems that are rooted in legal and ethical principles, in addition to requiring individual regulators to develop sector-

³² Worker Info Exchange (2023) Historic digital rights win for WIE and the ADCU over Uber and Ola at Amsterdam Court of Appeal. Available at: <https://www.workerinfoexchange.org/post/historic-digital-rights-win-for-wie-and-the-adcu-over-uber-and-ola-at-amsterdam-court-of-appeal>

³³ Ada Lovelace Institute (2021) *Regulate to innovate: a route to regulation that reflects the ambition of the UK AI Strategy*. Available at: <https://www.adalovelaceinstitute.org/report/regulate-innovate/>



specific codes of practice for the regulation of AI. While these recommendations are tailored toward the UK policy context, they convey some of our high level

Researchers looking to conduct research relating to platforms assessment of systemic risks/mitigation plans will need access to different categories of data. Examples of non-personal data/proprietary data that researchers should access include technical functionalities and relevant policies. For example, in our *Technical methods for regulatory inspection* report we identify analysis and research into measures such as systemic risk (under DSA) will require three kinds of evidence: **policies**, (e.g company hate speech policies), **processes** (e.g for recommending content – which may require interviews with members of staff) and **outcomes** (of policies, of algorithmic systems).